Final Project: Sensitivity Analysis using Linear Programming

UGST4090: Introduction to Optimization

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1 A Graphical Introduction to Sensitivity Analysis (Radioco Example)

Sensitivity analysis studies how changes in an LP's parameters affect the optimal solution.

Consider the Radioco radio production problem:

Radioco Radio Production Problem

Radioco manufactures two types of radios. The only scarce resource needed to produce radios is labor. The company now has two laborers. Laborer 1 is willing to work as many as 40 hours per week and is paid \$5 per hour. Laborer 2 is willing to work up to 50 hours per week and is paid \$6 per hour. The price as well as the resources required to build each type of radio are given in Table 1.

Table 1: Resources Required to Build Each Type of Radio

Radio	**Radio 1**	**Radio 2**
Price (\$)	25	22
Laborer 1:	1 hour	2 hours
Laborer 2:	2 hours	2 hours
Raw material cost	5	4

Let x_1 be the number of type 1 radios produced each week, and x_2 be the number of type 2 radios produced each week.

Each type of radio contributes the following profit:

- Type 1 radio: \$25
- Type 2 radio: \$22

The company must solve the following Linear Program (LP) to maximize profit:

$$\max z = 3x_1 + 2x_2$$

subject to the constraints:

$$x_1 + 2x_2 \leq 40 \quad \text{(Laborer 1 constraint)}$$

$$2x_1 + x_2 \leq 50 \quad \text{(Laborer 2 constraint)}$$

$$x_1, x_2 \geq 0 \quad \text{(Non-negativity constraint)}$$

Optimal Solution

Solving the system formed by the two binding constraints:

$$x_1 + 2x_2 = 40$$

$$2x_1 + x_2 = 50$$

gives:

$$x_1 = 20, \quad x_2 = 10$$

The corresponding profit is:

$$z = 3(20) + 2(10) = 80$$

Thus, the optimal solution is:

$$x_1 = 20, \quad x_2 = 10, \quad z = 80$$

At this solution, both labor constraints are binding, meaning that all available labor hours are used.

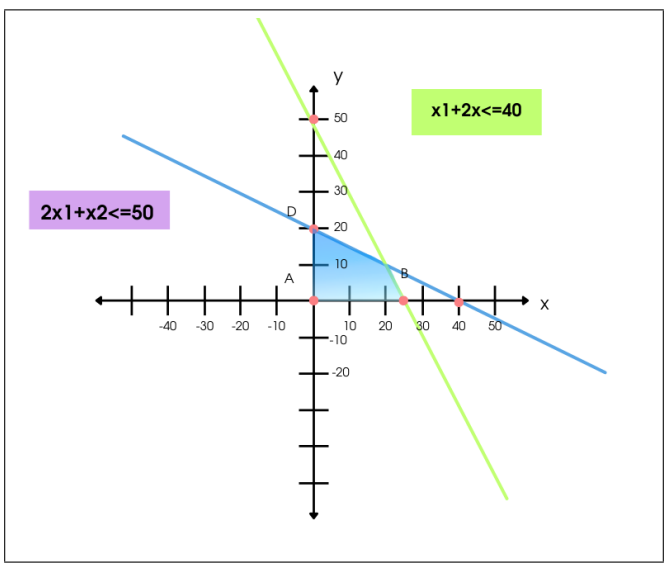

Figure 1: Visualizing Constraints and Feasible Region

1.1 Graphical Analysis of the Effect of a Change in an Objective Function Coefficient

Suppose the contribution to profit of type-1 radios changes from 3 to c_1.

The new objective function becomes:

$$\max z = c_1 x_1 + 2x_2$$

Each isoprofit line now has the form:

$$c_1 x_1 + 2x_2 = \text{constant}$$

or

$$x_2 = -\frac{c_1}{2} x_1 + \frac{\text{constant}}{2}$$

Thus, the slope of each isoprofit line is:

$$-\frac{c_1}{2}$$

Slopes of the Constraints

Laborer 1 constraint:

$$x_1 + 2x_2 = 40$$

$$x_2 = -\frac{1}{2} x_1 + 20$$

Slope:

$$-\frac{1}{2}$$

Laborer 2 constraint:

$$2x_1 + x_2 = 50$$

$$x_2 = -2x_1 + 50$$

Slope:

$$-2$$

For the current optimal basis to remain optimal, the slope of the isoprofit line must lie between the slopes of the two binding constraints.

Thus:

$$-2 \leq -\frac{c_1}{2} \leq -\frac{1}{2}$$

Multiplying by 2 and reversing inequalities gives:

$$4 \geq c_1 \geq 1$$

or:

$$1 \leq c_1 \leq 4$$

Interpretation

As long as:

$$1 \leq c_1 \leq 4$$

the optimal solution remains:

$$x_1 = 20, \quad x_2 = 10$$

If:

$$c_1 < 1$$

type-1 radios become relatively unprofitable and the optimal solution shifts toward producing more type-2 radios.

If:

$$c_1 > 4$$

type-1 radios become more profitable and the optimal solution shifts toward producing more type-1 radios.

—

1.2 Graphical Analysis of the Effect of a Change in a Right-Hand Side

Now suppose the number of hours available from Laborer 1 changes from 40 to:

$$40 + \Delta$$

The new constraint becomes:

$$x_1 + 2x_2 = 40 + \Delta$$

while the second constraint remains:

$$2x_1 + x_2 = 50$$

Solving these two equations gives:

$$x_1 = 20 + \frac{\Delta}{3}$$

$$x_2 = 10 - \frac{\Delta}{3}$$

Thus an increase in the available hours of Laborer 1 increases production of type-1 radios and decreases production of type-2 radios.

—

1.3 Shadow Prices

The shadow price of a constraint measures how much the optimal value of the objective function changes when the right-hand side of that constraint increases by one unit.

Substituting the expressions for x_1 and x_2 into the objective function gives:

$$z = 3x_1 + 2x_2$$

$$z = 3\left(20 + \frac{\Delta}{3}\right) + 2\left(10 - \frac{\Delta}{3}\right)$$

$$z = 80 + \frac{\Delta}{3}$$

Thus, increasing the available labor hours of Laborer 1 by one unit increases the optimal profit by:

$$\frac{1}{3}$$

Therefore, the shadow price of the first constraint is:

$$\frac{1}{3}$$

2 Sensitivity Analysis

Sensitivity analysis looks at a linear programming problem's optimal solution and asks the question: under what conditions does the optimal solution remain optimal? In other words, if the parameters of the problem change, we want to make sure that our solution is still optimal and does not have to be adjusted.

A linear programming solution is considered optimal if two conditions are met. First, the coefficients of the nonbasic variables in row 0 have to be nonnegative. This applies to a maximization problem. For a minimization problem all coefficients have to be nonpositive. The first condition ensures optimality. Second, the right-hand side (RHS) values of the constraints have to be nonnegative. The second condition ensures feasibility. If both conditions are satisfied, then the current basic feasible solution is optimal.

To perform sensitivity analysis, we need to follow two steps. First, we need to see how a change affects row 0 and the right-hand side values. Second, we check whether the feasibility and optimality conditions still hold. If the coefficients in row 0 and RHS values are still nonnegative, then the current solution is still optimal in a maximization problem. Otherwise, the current solution is not optimal.

A solution may become nonoptimal for two reasons. First, a negative coefficient could appear in row 0. This means that there is a higher optimum

possible with another basic feasible solution. This case is called a suboptimal basis. Second, a right-hand side value may become negative. In this case the current basis is no longer feasible, because one of the basic variables would have a negative value. This case is called an infeasible basis.

There are six types of changes possible to a linear programming problem, and we will review them in the following sections. We will discuss how these changes affect the optimal solution.

2.1 Changing the objective function coefficient of a nonbasic variable

In this case we change the coefficient in the objective function for a nonbasic variable. The solution remains feasible because the RHS values are not affected by the change. Since the variable is nonbasic , its value in the current solution is zero.

The current solution remains optimal if the coefficient is nonnegative. The value of the objective function does not change. If the coefficient is negative, then the solution becomes suboptimal. In this case the variable enters the set of basic variables, and we find a new solution, by solving the tableau with the new coefficient using the simplex method.

2.2 Changing the objective function coefficient of a basic variable

In this case we change the coefficient in the objective function for a basic variable. The solution remains feasible because the RHS values are not affected by the change. Since the variable is basic, we need to recompute the effects of the change for the entire row 0.

The current solution remains optimal if the coefficients in row 0 remain nonnegative. However, the optimal value changes, because the coefficient of the basic variable changes in the objective function.

If there are coefficients that are negative in row 0, we need to recalculate the problem as the current solution becomes suboptimal.

2.3 Changing the right-hand side of a constraint

Change in the RHS of a constraint would affect the optimal values of the decision variables and z value, and may affect the feasibility of the current basis.

To determine the effect on the basis, we recall that the RHS of the optimal

tableau can be expressed as:

$$\text{RHS} = B^{-1}b$$

where b is the vector of original RHS values and B^{-1} is the matrix of current values of original basis columns

Substituting $b + \Delta b$ for the modified RHS, the new tableau RHS becomes:

$$\text{RHS}_{\text{new}} = B^{-1}(b + \Delta b)$$

$$B^{-1}\begin{bmatrix} 48 \\ 20 + \Delta \\ 8 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -8 \\ 0 & 2 & -4 \\ 0 & -0.5 & 1.5 \end{bmatrix}\begin{bmatrix} 48 \\ 20 + \Delta \\ 8 \end{bmatrix}$$
$$= \begin{bmatrix} 24 + 2\Delta \\ 8 + 2\Delta \\ 2 - 0.5\Delta \end{bmatrix}$$

Figure 2: Effect of RHS perturbation on the optimal solution. From p. 243, *Operations Research: Applications and Algorithms.*

After computing RHS_{new}, two cases arise:

Case 1 — Basis becomes infeasible: if any entry of RHS_{new} is negative, the current basis is no longer feasible (some basic variable would take a negative value, violating $x \geq 0$). The dual simplex algorithm must be applied to recover a feasible optimal solution from the current basis.

Case 2 — Basis remains feasible: if all entries of RHS_{new} remain non-negative, the current basis remains both feasible and optimal. Also, reduced costs do not depend on b, so no re-checking of the objective row is required. The new decision variable values are given directly by RHS_{new}, and the new objective value is:

$$z_{\text{new}} = c_{BV} \cdot \text{RHS}_{\text{modify}} = c_{BV} \cdot B^{-1}(b + \Delta b)$$

where c_{BV} is the vector of objective coefficients of the current basic variables.

2.4 Changing the column of a nonbasic variable

In this situation we change several coefficients regarding a nonbasic variable at the same time. Multiple parameters are modified; however, the right-hand side values of the optimal solution remain unchanged; therefore, the current solution remains feasible.

The only value in row 0 that changes is the coefficient of the variable we are modifying. To check whether the current solution remains optimal we only need to check this value. If the coefficient is nonnegative, the current solution remains optimal. If the coefficient is negative, the solution becomes a suboptimal basis. In this case the variable can enter the set of basic variables and by

using the simplex algorithm we get a new optimal solution.

If we change the column of a basic variable the results would be difficult to determine because almost every part of the current solution would change. Therefore, it is usually faster to recalculate the optimal solution.

2.5 Adding a new activity

Another possibility is that we add a new variable to the solution. This variable would be regarded as a nonbasic variable, because it is currently not included in the solution.

When we add a new variable, the only change in the tableau is the newly added column and its coefficients. Therefore, to determine the optimality of the solution we only need to check if the row 0 coefficient of this variable is nonnegative.

If the coefficient is nonnegative in a maximization problem, then the current solution remains optimal. If the coefficient is negative, the variable can enter the set of basic variables and by using the simplex algorithm we get a new optimal solution, as in the previous case.

2.6 Adding a new constraint

There are three possible scenarios when we add a new constraint. The first scenario is that the current solution remains optimal. If we add a constraint to a linear programming problem, it either restricts the previous feasible region or does not affect it. If the current solution satisfies the new constraint, then the current solution remains optimal.

The second scenario is that the current solution is not optimal; however, there is a feasible solution. In this case we can use the dual simplex method. When using this method, we allow the right-hand side values to temporarily be negative. We pivot the tableau until the right-hand side values become nonnegative. In this algorithm we utilize the fact that row 0 satisfies the optimality condition. The problem can also be solved by the simplex algorithm; however, it might take multiple iterations.

The third scenario is that there is no feasible solution. In this case there is no solution that satisfies all constraints at the same time, which means that the linear programming problem is infeasible.

3 Sensitivity Analysis: Real-World Application for Price per Share Calculation

Sensitivity analysis is a tool used in finance to evaluate how changes in key assumptions impact financial outcomes, such as the price per share. This analysis is particularly useful in valuation models where uncertainty in inputs like discount rates (WACC) and terminal growth rates can significantly affect the final valuation of a company.

In the example provided,the sensitivity analysis demonstrates how variations in two key variables - WACC (Weighted Average Cost of Capital) and Terminal Growth Rate - influence the calculated price per share, which was initially calculated by Discounted Cash Flow (118.0) with Terminal growth = 1.25% and WACC = 10.63%.

3.1 Real-World Application

Sensitivity Analysis: Application

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Price Per Share	118.0
Terminal Growth	1.25%
WACC	10.63%

Sensitivity Analysis

	Terminal Growth					
WACC	118.0	1.00%	1.25%	1.50%	1.75%	2.00%
	9.50%	116.0	118.7	121.6	124.6	127.8
	10.00%	116.0	118.7	121.6	124.6	127.8
	10.50%	116.0	118.7	121.6	124.6	127.8
	11.00%	116.0	118.7	121.6	124.6	127.8
	11.50%	116.0	118.7	121.6	124.6	127.8

Figure 3: Sensitivity Analysis for Price per Share Calculation

- WACC (Discount Rate): Represents the cost of capital and is used to discount future cash flows. It varies based on market conditions and company risk.
- Terminal Growth Rate: Assumes the long-term growth rate of the company after the forecast period, which impacts the terminal value used in valuation.

Key Insights

- By adjusting these inputs, the model shows how sensitive the price per share is to changes in both WACC and Terminal Growth Rate.
- For example, a change in WACC from 9.5% to 11.5% might keep the price per share relatively stable at 116.0, but increasing the Terminal Growth Rate significantly can reduce the price to 106.4.

Summary

The sensitivity analysis presented here illustrates the impact of changes in WACC (Discount Rate) and Terminal Growth Rate on the price per share. By adjusting these key inputs, we see how the valuation fluctuates, with the price per share ranging from 106.4 to 127.8. This analysis helps assess the price sensitivity to these variables and provides insights into the potential risks and uncertainties in the valuation model.

References

[1] Winston, W. L. (2004). *Operations Research: Applications and Algorithms* (4th ed.). Thomson Brooks/Cole.

[2] "Linear Programming Sensitivity Analysis." YouTube, uploaded by [Channel Name], `https://youtu.be/N924D6tGOG8`